

Root Water Extraction and Limiting Soil Hydraulic Conditions Estimated by Numerical Simulation

Quirijn de Jong van Lier,* Klaas Metselaar, and Jos C. van Dam

ABSTRACT

Root density, soil hydraulic functions, and hydraulic head gradients play an important role in the determination of transpiration-rate-limiting soil water contents. We developed an implicit numerical root water extraction model to solve the Richards equation for the modeling of radial root water extraction. The average soil water content at the moment root water potential dropped below a defined critical value was then estimated. The dependence of average water content at the onset of plant water stress on potential transpiration and root density was compared with an analytical solution for hydraulic conditions in the root sphere. The critical value was a function of potential transpiration rate, soil hydraulic properties, and root density. Matric flux potential appears to be a convenient hydraulic property to determine the onset of limiting hydraulic conditions, as numerical simulations showed that, at onset, matric flux potential vs. distance from the root surface is independent of soil type. This was also determined analytically under the constant-rate assumption. Mean water content occurs at about 0.53 times the half-distance between roots. This allows calculation of the mean limiting soil water content and pressure head from the matric flux potential at this distance, which is a function of transpiration rate and root density only. A nomogram was developed that—given the transpiration rate, the root density, and the soil hydraulic functions—allows determination of the values of mean water content and mean pressure head that occur at the onset of transpiration reduction.

THE INTERACTION between crop transpiration rates and soil water availability is a complex issue. In an engineering approach used in irrigation scheduling and hydrologic modeling, a threshold value for water content (θ) or pressure head (h) is often assumed below which transpiration rates decrease below maximum values. Theoretical considerations (Gardner, 1960; Cowan, 1965) and experiments (Carbon, 1973; Zur et al., 1982) have established that root density, soil hydraulic functions, and hydraulic head gradients play an important role in the determination of this threshold value, and that the permanent wilting point (based on soil bulk values) may also vary as a function of these variables. At present, these factors are not considered in management or hydrological models. To illustrate this, Fig. 1 shows the widely used FAO irrigation guidelines for estimating limiting water content as a function of potential transpiration rate (Doorenbos and Kassam, 1986). Although four crop groups are defined and limiting water content is a

function of water content at field capacity and at the permanent wilting point, root characteristics and soil hydraulic properties are not considered in a direct way.

Modeling tools may provide insight as to how the interaction between soil and roots determines root water uptake. A great number of root water extraction models have been developed. Such models describe root water uptake based on the behavior of a single root (the microscopic approach) or describe uptake using properties of the overall root system (the macroscopic approach). Analytical solutions and numerical analyses for both types of models have been presented. Macroscopic models treat the root layer as a continuum of soil, water, and air with a sink (Gardner, 1964; Hillel et al., 1976; Hupet et al., 2002, 2003; Zuo and Zhang, 2002; Homae et al., 2002; Dardanelli et al., 2004; Raats, 2006). Although they are often more easily applicable to specific situations than microscopic models, they lack the capacity of a process-based estimation of the limiting θ or h value, which has to be estimated empirically. These models do not take into consideration the reduction of soil water content and, hence, the increase in hydraulic resistance close to water-extracting roots as confirmed experimentally by, e.g., Hainsworth and Aylmore (1986).

Microscopic models (e.g., Moldrup et al., 1992; Personne et al., 2003; Roose and Fowler, 2003; Raats, 2006) describe water extraction at an individual root level. The most common approach in these models is to consider the root as a straight cylindrical water sink, provoking a radial water flow toward it. This flow can be modeled with the Richards equation, which has been applied to solve numerous problems of unsaturated flow. A major constraint with respect to this equation is the highly nonlinear behavior of involved parameters, which limits analytical solutions to strict assumptions and boundary conditions. Analytical solutions for the microscopic approach have been presented by Gardner (1960) and Cowan (1965). A family of analytical solutions involves the quasilinear approximation (Pullan, 1990), but the advent of powerful computers now allows iterative numerical calculation techniques with less restrictive boundary conditions.

A mixed approach in which an analytical solution of the microscopic uptake profile has been combined with the overall behavior of the root system in a numerical three-dimensional model has been presented by Heinen (2001). At present, combined modeling of root architecture and flow patterns is envisaged (Darrach et al., 2006).

Explicit numerical models, in which flow processes are simulated in discrete steps in space and time, allow simulation of any kind of scenario but the resulting calculation procedures are usually very time consuming.

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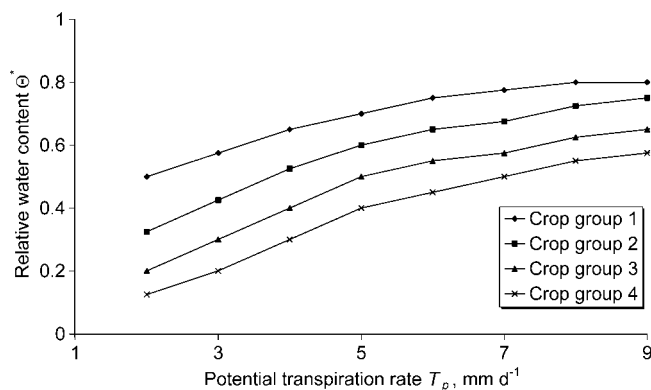


Fig. 1. Mean water content (θ) relative to field capacity (θ_{fc}) and the permanent wilting point (θ_{pwp}), $\Theta^* = (\theta - \theta_{pwp})/(\theta_{fc} - \theta_{pwp})$, at first occurrence of limiting hydraulic conditions, as a function of potential transpiration rate for four crop groups, as defined by Doorenbos and Kassam (1986).

Implicit schemes, in which a flow problem is described by a system of equations to be solved iteratively through time, may be adapted to many situations and require much less computer time to be processed. Van Dam and Feddes (2000) presented an implicit finite difference solution using the Richards equation for the simulation of infiltration, evaporation, and shallow groundwater levels. Their solution allows a flux-controlled or head-controlled surface boundary condition.

In this study, we developed an implicit numerical scheme that solves the Richards equation for radial root water extraction. The scheme uses an algorithm adapted from Van Dam and Feddes (2000). By defining a critical root water potential, the model was used to estimate the mean soil water content at the moment root water potential drops below the critical value. The dependence of this value on potential transpiration and root density was evaluated for three soils of the Dutch Staring series (Wösten et al., 2001) with different textures. An analytical solution for hydraulic conditions in the root sphere at first occurrence of water stress was used to support the conclusions. The overall objective was to derive critical parameter values for macroscopic root water extraction models from the microscopic (single-root) models.

MATERIAL AND METHODS

Soils and Hydraulic Properties

Root water extraction and limiting hydraulic conditions were analyzed using hydraulic data for three topsoils from the Dutch Staring series (Wösten et al., 2001) as listed in Table 1. For these soils, K - θ - h relations can be described by the van Genuchten (1980) equation system:

$$\Theta = [1 + |\alpha h|^n]^{(1/n)-1} \quad [1]$$

$$K = K_{sat} \Theta^\lambda [1 - (1 - \Theta^{n/(n-1)})^{1-(1/n)}]^2 \quad [2]$$

in which $\Theta = (\theta - \theta_r)/(\theta_s - \theta_r)$; θ , θ_r , and θ_s are water content, residual water content and saturated water content ($\text{m}^3 \text{m}^{-3}$), respectively; h is pressure head (m); K and K_{sat} are hydraulic conductivity and saturated hydraulic conductivity, respectively (m d^{-1}); and α (m^{-1}), n , and λ are empirical parameters.

Table 1. Van Genuchten parameters (Eq. [1] and [2])† for three soils from the Dutch Staring series used in simulations.

Staring soil ID	Textural class	Reference in this paper	θ_r	θ_s	K_{sat}	α	λ	n
			$\text{m}^3 \text{m}^{-3}$	$\text{m}^3 \text{m}^{-3}$	m d^{-1}	m^{-1}		
B3	Loamy sand	Sand	0.02	0.46	0.1542	1.44	-0.215	1.534
B11	Heavy clay	Clay	0.01	0.59	0.0453	1.95	-5.901	1.109
B13	Sandy loam	Loam	0.01	0.42	0.1298	0.84	-1.497	1.441

† θ_r , residual volumetric soil water content; θ_s , saturated volumetric soil water content; K_{sat} , saturated hydraulic conductivity; α , λ , and n are empirical parameters.

A convenient soil hydraulic property that is often used in soil water movement studies (e.g., Raats, 1977; Warrick and Amoozegar-Fard, 1977; Pullan, 1990) and will be used in this study is the matric flux potential M_{h_0} ($\text{m}^2 \text{d}^{-1}$), the integral of hydraulic conductivity over pressure head starting at the pressure head at permanent wilting point:

$$M_{h_0} = \int_{h_0}^h K(h) dh \quad [3]$$

Soil water content, hydraulic conductivity, and matric flux potential (integration started at $h_0 = -150 \text{ m}$) as a function of hydraulic head are shown for the three soils in Fig. 2.

Implicit Numerical Modeling

Darcy's equation combined with the mass conservation principle leads to the Richards equation, which for one-dimensional axisymmetric flow is written as

$$\begin{aligned} \frac{\partial \theta}{\partial t} &= C(h) \frac{\partial h}{\partial t} = -\frac{q}{r} - \frac{\partial q}{\partial r} - S = -\frac{\partial(rq)}{r \partial r} - S \\ &= \frac{1}{r} \frac{\partial}{\partial r} \left[r K(h, \theta) \frac{\partial H}{\partial r} \right] - S \end{aligned} \quad [4]$$

where t is time (d), C is the differential water capacity ($d\theta/dh$, m^{-1}), q is the water flux density (m d^{-1}), r is the distance from the axial center (m), S is a sink or source term (d^{-1}), and H is the hydraulic head (m). To solve the Richards equation for the case of radial root water extraction, a numerical scheme based on the scheme presented by Van Dam and Feddes (2000) was used with the following modifications:

- the gravitational component is not considered in the root extraction model
- fluxes are adapted to the one-dimensional axisymmetric nature of root extraction, i.e., fluxes converge as they approach the root, proportional to the distance from the axial center
- there is no sink (the only water exit is the root surface located at the inner side of the first compartment)
- flux density at the outermost compartment is set to zero
- flux density at the innermost compartment (the one immediately neighboring the root) is set equal to flux density entering the root, which is determined by transpiration rate and total root area
- if the pressure head in the innermost compartment drops below a critical value h_{lim} (in this study $h_{lim} = -150 \text{ m}$), pressure head is fixed to this critical value and the flux density adjusted to that

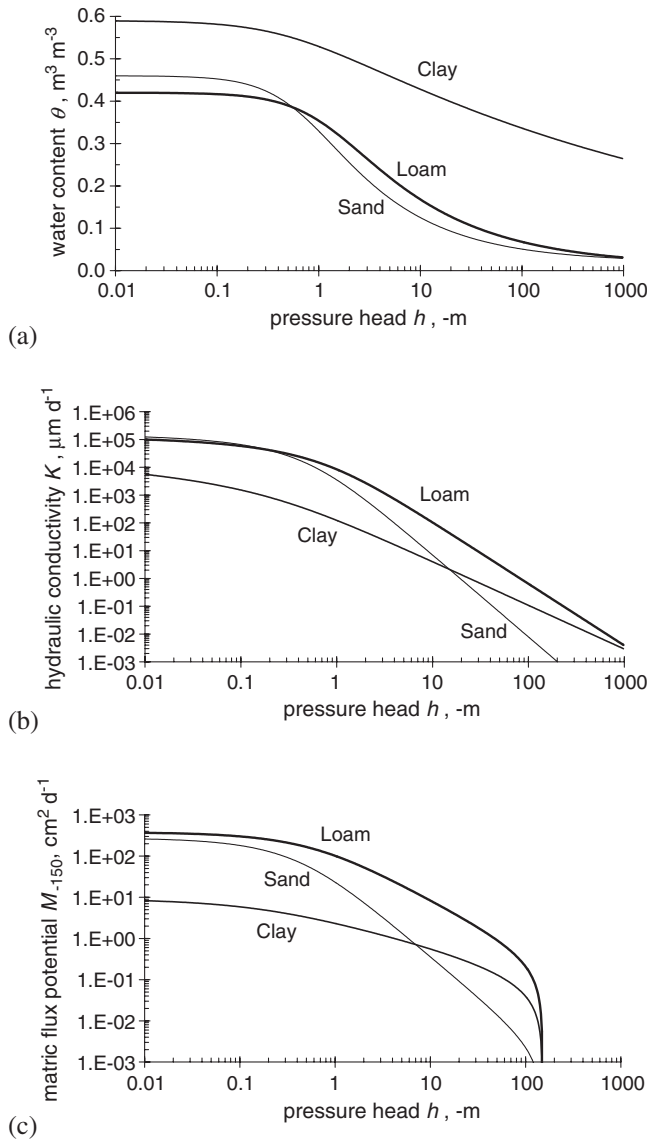


Fig. 2. (a) Soil water content, (b) hydraulic conductivity, and (c) matric flux potential as a function of pressure head for sand, clay, and loam soils.

Segment size (d_r) was chosen smaller near the root and larger at greater distance, according to

$$d_r = d_{r,\min} + (d_{r,\max} - d_{r,\min}) \left(\frac{r - r_0}{r_m - r_0} \right)^S \quad [5]$$

with $d_{r,\min} = 10^{-8}$ m, $d_{r,\max} = 5.10^{-4}$ m, $S = 0.5$, r_0 (m) is the root radius and r_m (m) is the radius of the root extraction zone, equal to the half-distance between roots (r_m), which relates to the root density R (m m^{-3}) as

$$R = \frac{1}{\pi r_m^2} \quad [6]$$

It can also be shown that

$$R = \frac{L}{z A_p} \quad [7]$$

and

$$L = \frac{A_r}{2\pi r_0} \quad [8]$$

in which L (m) is the root length, z (m) is the total rooted soil depth, A_p (m^2) is the surface area and A_r (m^2) is the root surface area. Combining these equations, it follows that

$$r_m = \sqrt{\frac{A_p z}{\pi L}} = \sqrt{\frac{2 A_p z r_0}{A_r}} \rightarrow \frac{A_p}{A_r} = \frac{r_m^2}{2 z r_0} \quad [9]$$

Figure 3 shows a schematic representation of the root and axial segments considered for the simulations, including r , r_0 and r_m . The chosen segment size distribution resulted in 24, 73, and 228 segments for the high, medium, and low root density simulations, respectively. Further reduction of the segment size did not lead to significant differences in simulation results. In addition, intermediate K values were calculated using arithmetic or harmonic means, but no significant difference was found, confirming the chosen space step to be sufficiently small. Time steps were automatically adapted by the computer program to reach convergence of the iterative matrix solution in three to five iterations. In this way, observed time steps ranged from 1 s to 1 h.

Assuming no sink or source ($S = 0$) and no gravitational or osmotic component ($H = h$), Eq. [4] reduces to

$$C(h) \frac{\partial h}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left[r K(h, \theta) \frac{\partial h}{\partial r} \right] \quad [10]$$

Equation [10] can be discretized by explicit linearization of K and C :

$$\frac{h_i^{j+1} - h_i^j}{\Delta t^j} = \frac{r_{i-0.5} K_{i-0.5}^j \frac{\Delta h_{i-0.5}^{j+1}}{r_i - r_{i-1}} - r_{i+0.5} K_{i+0.5}^j \frac{\Delta h_{i+0.5}^{j+1}}{r_{i+1} - r_i}}{r_i C_i^j \Delta r_i} \quad [11]$$

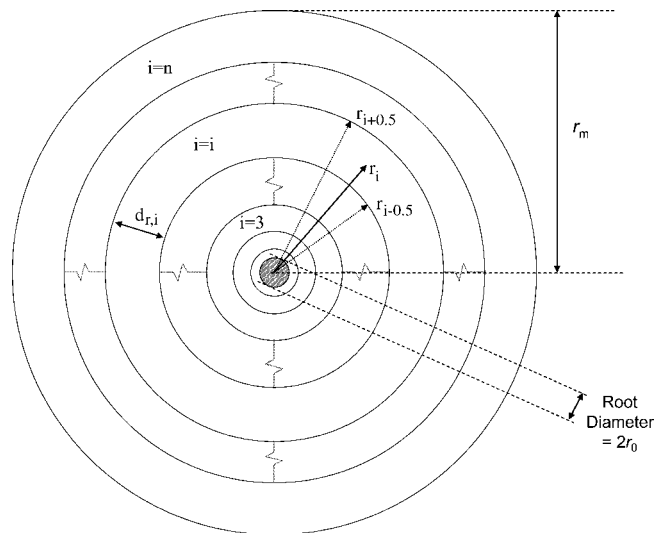


Fig. 3. Schematic representation of root and axial segments as used for the simulations; r_m is the half-distance between roots, r_0 is the root radius, $d_{r,i}$ is the segment size for segment i , and r_i is the distance from the axial center for segment i .

where i is the segment number (Segment 1 closest to the root) and j is the time step. In this equation, indexes refer to axial segments as illustrated in Fig. 3.

Analogous to Van Dam and Feddes (2000), Eq. [10] can be discretized using the finite difference scheme of Eq. [11], yielding the following equation to be solved iteratively:

$$C_i^{j+1,p-1}(h_i^{j+1,p} - h_i^{j+1,p-1}) + \theta_i^{j+1,p-1} - \theta_i^j = \frac{t^{j+1} - t^j}{r_i \Delta r_i} \left[r_{i-0.5} K_{i-0.5}^j \frac{h_{i-1}^{j+1,p} - h_{i-1}^{j+1,p-1}}{r_i - r_{i-1}} - r_{i+0.5} K_{i+0.5}^j \frac{h_{i+1}^{j+1,p} - h_{i+1}^{j+1,p-1}}{r_{i+1} - r_i} \right] \quad [12]$$

where p is the iteration level.

A solution to the problem can be found by applying Eq. [12] to each segment and solving the resulting equation system described by a tridiagonal matrix as follows:

$$\begin{bmatrix} \beta_1 & \gamma_1 & & & \\ \alpha_2 & \beta_2 & \gamma_2 & & \\ & \alpha_3 & \beta_3 & \gamma_3 & \\ & & & \alpha_{n-1} & \beta_{n-1} & \gamma_{n-1} \\ & & & & \alpha_n & \beta_n \end{bmatrix} \begin{bmatrix} h_1^{j+1,p} \\ h_2^{j+1,p} \\ h_3^{j+1,p} \\ h_{n-1}^{j+1,p} \\ h_n^{j+1,p} \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_{n-1} \\ f_n \end{bmatrix}$$

in which α_i , β_i , γ_i , and f_i are defined as follows:

1. Intermediate segments ($i = 2$ to $i = n - 1$):

$$\alpha_i = -\frac{t^{j+1} - t^j}{r_i(r_i - r_{i-1})\Delta r_i} r_{i-0.5} K_{i-0.5}^j \quad [13]$$

$$\beta_i = C_i^{j+1,p-1} + \frac{t^{j+1} - t^j}{r_i(r_i - r_{i-1})\Delta r_i} r_{i-0.5} K_{i-0.5}^j + \frac{t^{j+1} - t^j}{r_i(r_{i+1} - r_i)\Delta r_i} r_{i+0.5} K_{i+0.5}^j \quad [14]$$

$$\gamma_i = -\frac{t^{j+1} - t^j}{r_i(r_{i+1} - r_i)\Delta r_i} r_{i+0.5} K_{i+0.5}^j \quad [15]$$

$$f_i = C_i^{j+1,p-1} h_i^{j+1,p-1} - \theta_i^{j+1,p-1} + \theta_i^j \quad [16]$$

- 2A. Root segment ($i = 1$) with flux boundary condition (applied while $h_1 > h_{lim}$, i.e., when transpiration meets potential transpiration demand):

$$\beta_1 = C_1^{j+1,p-1} + \frac{t^{j+1} - t^j}{r_1(r_2 - r_1)\Delta r_1} r_{1.5} K_{1.5}^j \quad [17]$$

$$\gamma_1 = -\frac{t^{j+1} - t^j}{r_1(r_2 - r_1)\Delta r_1} r_{1.5} K_{1.5}^j \quad [18]$$

$$f_1 = C_1^{j+1,p-1} h_1^{j+1,p-1} - \theta_1^{j+1,p-1} + \theta_1^j - \frac{r_0 q_{root}(t^{j+1} - t^j)}{r_1 \Delta r_1} \quad [19]$$

in which $q_{root} = T_p A_p / A_r$, A_p being the surface area (m^2) and A_r the root area (m^2).

- 2B. Root segment ($i = 1$) with head boundary condition (applied while $h_1 = h_{lim}$, i.e., when transpiration is less than potential transpiration demand):

$$\beta_i = C_1^{j+1,p-1} + \frac{t^{j+1} - t^j}{r_1(r_1 - r_0)\Delta r_1} r_{0.5} K_{0.5}^j + \frac{t^{j+1} - t^j}{r_1(r_2 - r_1)\Delta r_1} r_{1.5} K_{1.5}^j \quad [20]$$

$$\gamma_1 = -\frac{t^{j+1} - t^j}{r_1(r_2 - r_1)\Delta r_1} r_{1.5} K_{1.5}^j \quad [21]$$

$$f_1 = C_1^{j+1,p-1} h_1^{j+1,p-1} - \theta_1^{j+1,p-1} + \theta_1^j + \frac{(t^{j+1} - t^j)}{r_1(r_1 - r_0)\Delta r_1} r_{0.5} K_{0.5}^j h_{lim} \quad [22]$$

3. Outer border segment ($i = n$):

$$\alpha_n = -\frac{t^{j+1} - t^j}{r_n(r_n - r_{n-1})\Delta r_n} r_{n-0.5} K_{n-0.5}^j \quad [23]$$

$$\beta_n = C_n^{j+1,p-1} + \frac{t^{j+1} - t^j}{r_n(r_n - r_{n-1})\Delta r_n} r_{n-0.5} K_{n-0.5}^j \quad [24]$$

$$f_n = C_n^{j+1,p-1} h_n^{j+1,p-1} - \theta_n^{j+1,p-1} + \theta_n^j \quad [25]$$

Intermediate K values (at positions $i - 0.5$ or $i + 0.5$) were calculated from K values at positions $i - 1$, i , and $i + 1$ obtained by Eq. [2] using the geometric mean.

Relations between Bulk Properties and Microscopic Properties

Mean water content (θ_{mean}) was calculated as the weighted average of all segment water contents:

$$\theta_{mean} = \frac{\int_{r_0}^{r_m} \theta(r) r dr}{\int_{r_0}^{r_m} r dr} \quad [26]$$

Mean pressure head (h_{mean}), mean hydraulic conductivity (K_{mean}) and mean matric flux potential (M_{mean}) were calculated from the mean water content by Eq. [1] to [3]:

$$\begin{aligned} h_{mean} &= h(\theta_{mean}) \\ K_{mean} &= K(\theta_{mean}) \\ M_{mean} &= M(\theta_{mean}) \end{aligned} \quad [27]$$

Transpiration demand, a function of the atmospheric conditions, together with soil hydraulic state variables, determines pressure head at the root surface. Its value, however, is limited to a minimum, h_{lim} , here assumed to be equal to the commonly used pressure head at the permanent wilting point, -150 m. The instant of first occurrence of h_{lim} at the root surface is very important, as from that moment on, water root uptake no longer meets transpiration demand. As depletion continues, relative transpiration will decrease and stop completely when the pressure head has dropped to the permanent wilting point everywhere in the soil cylinder surrounding the root. When the root potential reaches h_{lim} for the first time, h_{mean} as calculated by Eq. [26] and [27] is not equal to h_{lim} but, instead,

has a higher value that depends on the entire root extraction history. We will refer to the hydraulic conditions at this instant as limiting hydraulic conditions, which will be defined by the subscript mean,lim, as in $h_{\text{mean,lim}}$. Thus, $h_{\text{mean,lim}}$ is h_{mean} at the first occurrence of h_{lim} at the root surface. The determination of $h_{\text{mean,lim}}$ and $\theta_{\text{mean,lim}}$ is a major challenge in soil physics. Simulation of flux phenomena in the root zone can help in doing so, as we will explain.

Simulation Scenarios

The main simulation scenario parameters are listed in Table 2. The variation in root densities ($0.01\text{--}1\text{ m m}^{-3}$) and transpiration rates ($3\text{--}6\text{ mm d}^{-1}$) was chosen in such a way that main crop types and climatic conditions are included (see, e.g., Stalham and Allen [2001], Pietola and Alakukku [2005], and Gravel et al. [2002] on root density, and Allen et al. [1998] on transpiration rates). Potential transpiration (T_p , mm d^{-1}) was supposed to vary daily according to a sine-wave function with its amplitude equal to its mean value $T_{p,\text{mean}}$. In this way, during a 24-h period, the transpiration rate varied between a minimum of zero and a maximum of $2T_{p,\text{mean}}$. In the simulations, we assumed pressure head at the root surface to be variable, but not able to fall below -150 m . The value of -150 m is assumed to be independent of all other parameters.

The scenarios were set up to illustrate (i) how pressure head at the root surface, mean soil pressure head, and transpiration rate behave with time for different combinations of potential transpiration and root density; and (ii) how mean soil hydraulic conditions (pressure head, hydraulic conductivity, matric flux potential, and water content) vary according to soil type, potential transpiration rate, and root density at first occurrence of a limiting pressure head at the root surface.

RESULTS AND DISCUSSION

Root Pressure Head, Mean Pressure Head, and Transpiration as a Function of Time

Simulation results are shown in Fig. 4 for the two potential transpiration rates and three root densities for the clay soil. The daily fluctuation of the pressure head in the root (h_{root}) to match varying transpiration can be ob-

served in this figure, especially at low and medium R . For the low R , within a few days h_{root} reaches much more negative values than h_{mean} , with a minimum every 24 h simultaneous with the maximum in the daily transpiration curve. In contrast, at high R , h_{root} remains almost equal to h_{mean} from beginning to end of the simulations.

This difference between low and high R is caused by two things. In the first place, at high R , with total root area equal to 100 times the total root area at low R , flux densities toward the root need to be only 1/100 of their values at low R . Consequently, hydraulic gradients will be much smaller at high than at low R , and there will be less difference between values of pressure head at the root surface and in the surrounding soil. Second, at low R , distances between neighboring roots are on the order of 10 cm, while in a high R scenario this distance is only about 1 cm. This allows much greater differences to occur between pressure heads at the root surface and at the outer side of the root influence sphere. This effect is clearly observed when comparing Fig. 4a with 4c, but even more pronounced at higher T_p (Fig. 4d with 4f). In Fig. 5, the same can be seen from another point of view, showing pressure head as a function of distance from the axial center at the moment of first occurrence of limiting root water potential. In this figure, it can be seen that pressure head gradients need to be much higher at the lower R . As a consequence, the difference between mean and root pressure head increases.

At medium root density (Fig. 4b and 4e), during the first days when water content is still high, hydraulic conductivity is sufficiently high to allow low pressure head gradients, resulting in relatively small differences between h_{root} and h_{mean} . At a certain moment, water content and hydraulic conductivity become small and higher gradients are necessary during the hours of the day with high transpiration. From that moment on, the daily fluctuation of h_{root} is similar to that at low R (Fig. 4a and 4d).

After the first occurrence of h_{lim} , root water uptake no longer meets the potential transpiration demand, as shown by the simultaneous reduction in relative transpiration in all simulations (cf. Fig. 4). Comparing the three root densities at one transpiration level, it is clear that a higher R increases the time to first occurrence of limiting root water potential. For the case of low T_p (Fig. 4a–4c) it takes 14, 31, and 34 d at low, medium, and high R , respectively; at high T_p (Fig. 4d–4f) this changes to 3, 14, and 17 d, respectively. Therefore, in this soil and as far as water uptake is concerned, there is a great benefit for a crop in increasing R from low to medium, whereas the gain of an increase from medium to high R is small.

These results depend on the actual soil hydraulic functions, although the overall tendency is not expected to change: higher transpiration demand and lower root density cause greater discrepancy between h_{root} and h_{mean} and, therefore, limiting hydraulic conditions are reached at higher values of h_{mean} and θ_{mean} . Low hydraulic conductivity in the root zone water content range will cause greater discrepancy between h_{root} and h_{mean} . Under these conditions, root density becomes more important for root water extraction than in soils with a higher conductivity.

Table 2. Parameters for simulation scenarios.

Description	Symbol	Scenario	Value	Unit
Plant surface area	A_p		0.04	m^2
Rooting depth	z		0.5	m
Limiting root potential	h_{lim}		-150	m
Root radius	r_0		0.5	mm
Root length	L	low R	2	m
		medium R	20	m
		high R	200	m
Root density	$R = L/zA_p$	low R	0.01	m m^{-3}
		medium R	0.1	m m^{-3}
		high R	1	m m^{-3}
Half distance between roots	$r_m = \sqrt{A_p z / \pi L}$	low R	56.5	mm
		medium R	17.8	mm
		high R	5.65	mm
Initial pressure head	h_{ini}		-1	m
Mean potential transpiration rate†	T_p	low T_p	3	mm d^{-1}
		high T_p	6	mm d^{-1}
Daily amplitude in potential transpiration rate†	a_T	low T_p	3	mm d^{-1}
		high T_p	6	mm d^{-1}
Soil type		sand	Table 1	
		clay	Table 1	
		loam	Table 1	

† For the determination of limiting hydraulic conditions, $2T_p$ and zero amplitude were used.

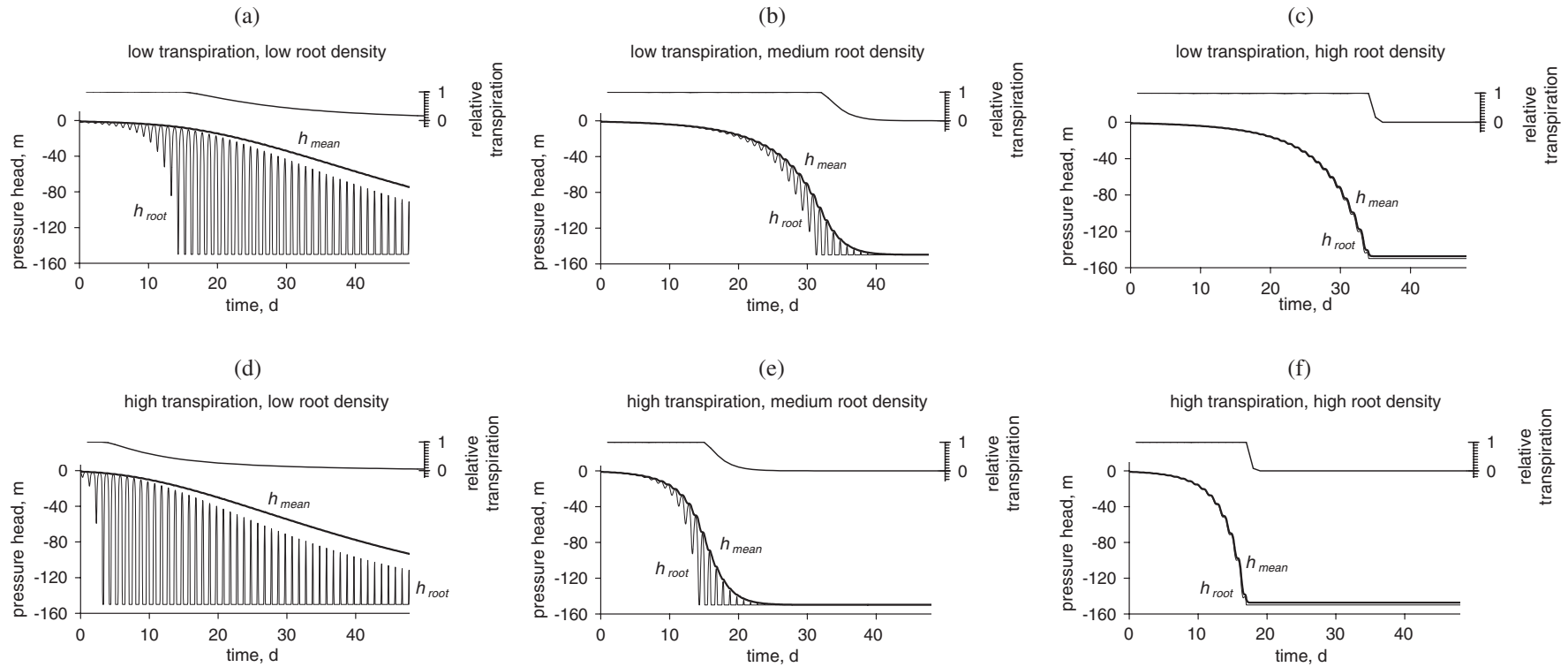


Fig. 4. Simulated pressure head at the root surface (h_{root}), mean pressure head (h_{mean}) and relative transpiration as a function of time for low and high potential transpiration rates and low, medium, and high root density in a clay soil.

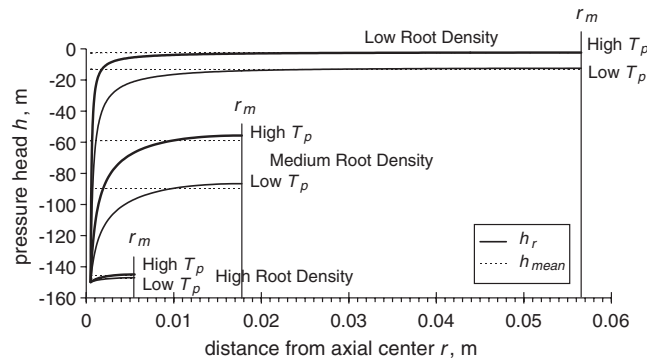


Fig. 5. Pressure head as a function of distance from the axial center (h_r) and mean pressure head (h_{mean}) for low and high potential transpiration rates (T_p) for simulations with low, medium, and high root density, at the moment of first occurrence of limiting root water potential in clay soil; r_m is the radius of the root extraction zone or the half-distance between roots.

Once the limiting conditions are reached, relative transpiration decreases. In the low- R scenarios, at this moment much water is still available farther away from the roots, as can also be seen in Fig. 5. During the night, transpiration decreases but established pressure head gradients maintain water movement toward the root, increasing the water content near the root. This results in a temporary relief of water stress: root pressure head becomes less negative and, during the first part of the next day, extraction can cope with potential transpiration. Due to this effect, relative transpiration decreases more slowly in the low- R scenarios. In contrast, in high- R scenarios, the exploited soil volume is almost completely depleted at the first occurrence of limiting pressure head at the root surface. As a consequence, the transpiration rate decreases much more abruptly from its maximum value to zero.

Dependence on Transpiration Rate: Diurnal Fluctuation vs. Constant Rate

To analyze the dependence of $h_{\text{mean,lim}}$ and the corresponding $\theta_{\text{mean,lim}}$ on potential transpiration rates, a series of simulations were performed with the three soils and different root densities.

Results for the clay soil at transpiration rates from 2 to 8 mm d^{-1} are shown by the sawtooth curves identified with A in Fig. 6. The variable shape of these lines is caused by the fact that daily variations in transpiration are simulated: when root water potential approaches its limiting value at the beginning of a night period with low potential transpiration, the occurrence of the limiting value is postponed to the next day, consequently allowing an additional small reduction in water content and pressure head. On the other hand, if the limiting root potential is reached simultaneous with a high daytime transpiration, the first occurrence of water stress will tend to be sooner and at less negative h_{mean} .

If daily amplitude is disregarded, simulations result in the smooth lines marked B in Fig. 6. The associated estimates of $h_{\text{mean,lim}}$ are less realistic for not considering the variation in daily transpiration. To avoid underestimation of $h_{\text{mean,lim}}$, the most secure estimate of

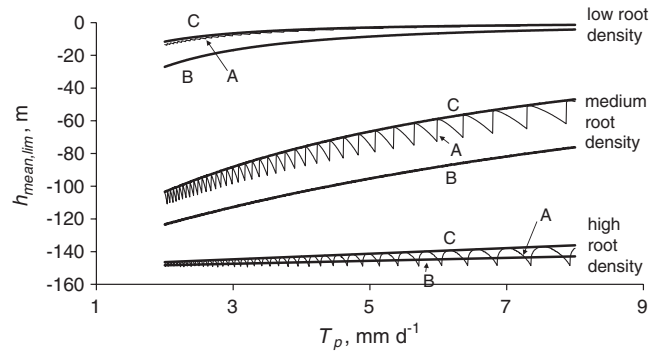


Fig. 6. Mean pressure head at first occurrence of limiting hydraulic conditions ($h_{\text{mean,lim}}$) as a function of potential transpiration rate (T_p) for low, medium, and high root density in a clay soil simulated with and without daily transpiration amplitude (for lines identified with "A", mean transpiration equals T_p and amplitude equals T_p ; for lines identified with "B", mean transpiration equals T_p and amplitude is zero; for lines identified with "C", mean transpiration equals $2T_p$ and amplitude is zero).

$h_{\text{mean,lim}}$ as a function of transpiration rate would be given by a line connecting the maxima in the sawtooth A lines. This can be achieved, in good approximation, using double the transpiration rate with zero amplitude, as shown by the lines marked C in Fig. 6 that refer to simulations with a constant transpiration rate twice as high as the axis labels indicate. This confirms that the local maxima in A correspond to the highest daily transpiration rates, equal to $2T_p$. Subsequent simulations were performed at a constant level equal to $2T_p$, corresponding to lines C. Hence, when referring to $h_{\text{mean,lim}}$ at transpiration rate T_p considering daily amplitude, the corresponding simulation was run at $2T_p$ without daily amplitude.

Selection of a Key Soil Physical Variable Controlling Transpiration Rate

Macroscopic limiting soil water conditions are often defined with respect to some spatial average of h (e.g., Feddes et al., 1988) or some spatial average of θ (e.g., Doorenbos and Kassam, 1986). Equating these spatial averages with the mean values calculated using the numerical results, Fig. 7 and 8 show that these mean values are a function of transpiration rate, soil type, and root density. A function relating relative transpiration to matric head or soil water content should then include transpiration rate, soil type (soil hydraulic parameters), and root density as parameters. Comparing Fig. 8 with the FAO guidelines (Doorenbos and Kassam, 1986) for estimating limiting water content depicted in Fig. 1 shows similarity in order of magnitude between the FAO guidelines and the simulated low root density. The FAO approach, however, does not include soil physical factors.

Hydraulic conductivity is a key parameter in soil water movement and therefore it might be an alternative indicator of limiting hydraulic conditions. Figure 9 shows K values corresponding to $h_{\text{mean,lim}}$ as a function of transpiration rate for the three root densities and soils. Although values are relatively similar, especially for medium to high root densities and low transpiration

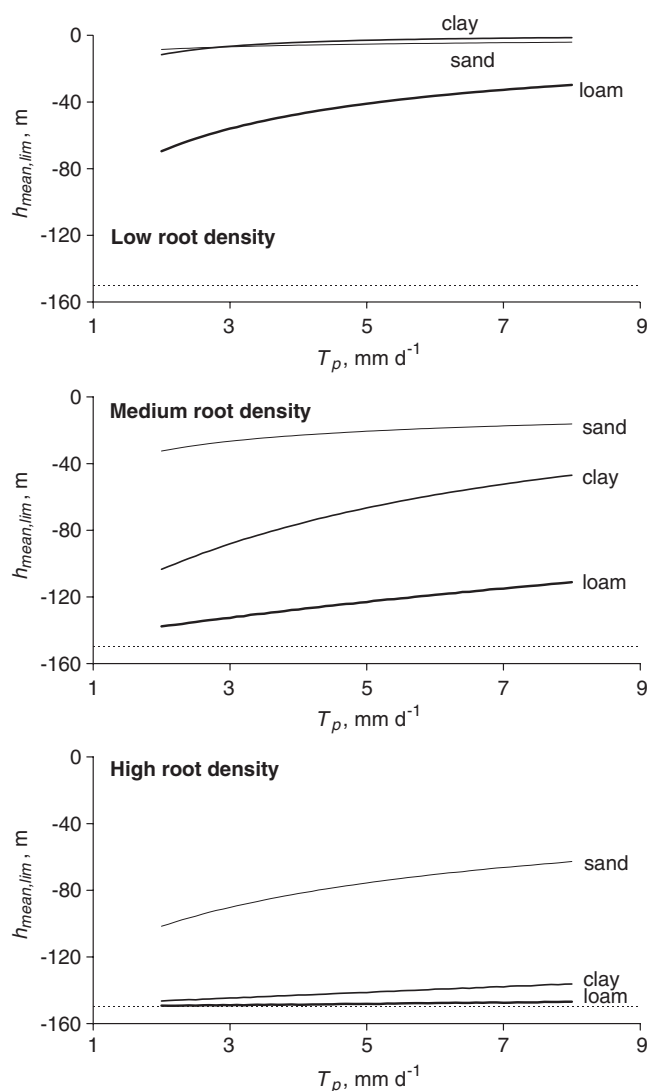


Fig. 7. Mean pressure head at first occurrence of limiting hydraulic conditions ($h_{\text{mean,lim}}$) as a function of potential transpiration rate (T_p) for low, medium, and high root density in the sand, loam and clay soils (dotted lines refer to minimum root pressure head $h = -150$ m).

rates, hydraulic conductivity does not seem to be a better indicator than h or θ for limiting hydraulic conditions.

Given its definition (Eq. [3]), matric flux potential (M) integrates hydraulic conductivity and pressure head, providing another option for analysis. An analytical expression for M is not straightforward using the Mualem–van Genuchten equation system (Eq. [1] and [2]). Therefore $M(h)$ was calculated by numerical integration of K , choosing $h_0 = h_{\text{pwp}}$ (h at the permanent wilting point) = -150 m. For h less than -100 m, the interval was subdivided in 1,000 equal steps; for h between -100 and -30 m, this number was 10,000; between -30 and -10 m the number of steps was 30,000; and for h greater than -10 m, the number of steps was 100,000. Evaluating the value of M corresponding to $h_{\text{mean,lim}}$, the result is a linear relation between M and T_p with intercept equal to zero, the slope of which is a function of root density, but independent of soil type (Fig. 10). Based on this result, matric flux potential M is a convenient parameter

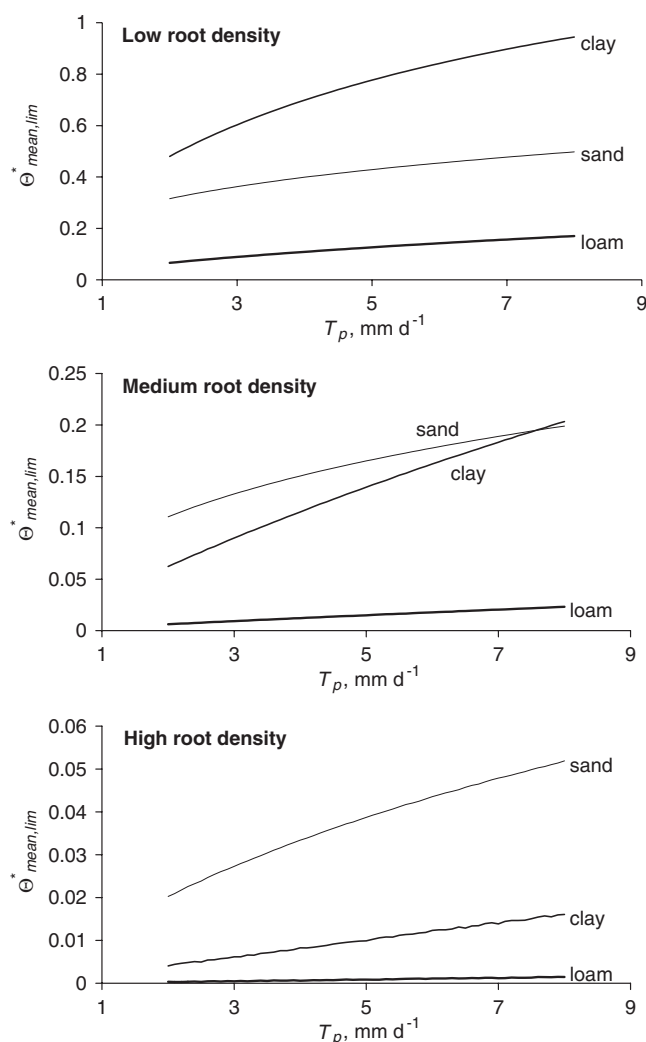


Fig. 8. Mean water content (θ) relative to field capacity (θ_{fc}) and the permanent wilting point (θ_{pwp}), $\Theta^* = (\theta - \theta_{\text{pwp}})/(\theta_{\text{fc}} - \theta_{\text{pwp}})$, at first occurrence of limiting hydraulic conditions ($\Theta^*_{\text{mean,lim}}$) as a function of potential transpiration rate (T_p) for low, medium, and high root density in the sand, loam, and clay soils.

to define limiting hydraulic conditions. Disregarding hysteresis, M is a strictly monotonic function of h , allowing conversion between M and h or θ . Limiting hydraulic conditions can be characterized by a single M value, independent of soil type, due to the fact that M combines both factors that determine water movement, hydraulic conductivity K and pressure head difference dh . Water flux toward roots is then only defined by transpiration rate and root density.

The ratio between $M_{\text{mean,lim}}$ and T_p was determined for several half-distances between roots ranging from 5.65 to 56.5 mm, corresponding to the range between high and low root density in this study. A power function showed a good fit to the results (Fig. 11), with (for M/T_p and r_m expressed in meters):

$$\frac{M_{\text{mean,lim}}}{T_p} = p r_m^q \quad [28]$$

with $p = 23.5$ and $q = 2.367$ ($n = 7$, $r^2 = 0.9997$).

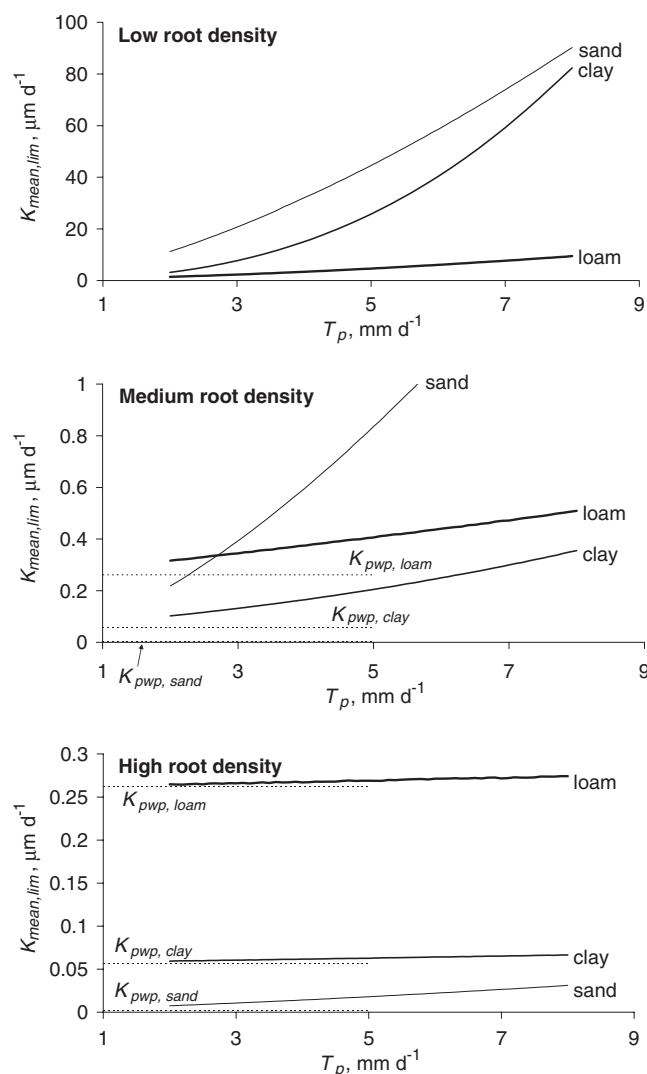


Fig. 9. Hydraulic conductivity at first occurrence of limiting hydraulic conditions ($K_{\text{mean,lim}}$) as a function of potential transpiration rate (T_p) for low, medium, and high root density in the sand, loam and clay soils. (K_{pwp} is the hydraulic conductivity at the permanent wilting point for the respective soil).

A Nomogram to Estimate Soil Water State at First Occurrence of Limiting Hydraulic Conditions

For any given root density and transpiration rate, Eq. [28] can be used to calculate the corresponding $M_{\text{mean,lim}}$, which then can be converted to $\theta_{\text{mean,lim}}$ or $h_{\text{mean,lim}}$ by numeric means using the van Genuchten–Mualem Eq. [1] and [2]. Using this result, a nomogram can be designed as shown in Fig. 12, allowing determination of $h_{\text{mean,lim}}$ and $\theta_{\text{mean,lim}}$ if potential transpiration and root density are known. Dotted arrows illustrate nomogram use for a transpiration rate of 6 mm d^{-1} and a medium root density (0.1 m m^{-3}) for the sand soil. In this case, $\theta_{\text{mean,lim}}$ corresponds to 0.17 and $h_{\text{mean,lim}}$ is shown to be close to -20 m , in agreement with the corresponding value from Fig. 7.

Alternatively, the nomogram can be used to determine which combination of root density and potential transpiration results in a $h_{\text{mean,lim}}$ of a certain magnitude.

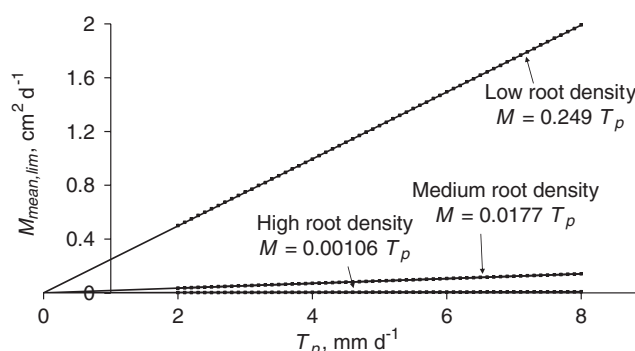


Fig. 10. Matric flux potential at first occurrence of limiting hydraulic conditions ($M_{\text{mean,lim}}$) as a function of potential transpiration rate (T_p) for low, medium, and high root density. Lines for all three soils are coincident.

For example, choosing $h_{\text{mean,lim}} = -10 \text{ m}$ and $T_p = 6 \text{ mm d}^{-1}$, it can be read from the nomogram that the corresponding root density is between 0.02 and 0.05 m m^{-3} (i.e., in the range between low and medium R) for the sand and clay soils, and between 0.002 and 0.005 m m^{-3} (i.e., smaller than low R) for the loam soil. This is in agreement with Fig. 7 in which all represented values (except for clay and sand in combination with low R) are below -10 m . As water stress in field crops is commonly observed at values less negative than -10 m , while root densities are reported in the range of medium to high R , the values from Fig. 7 are lower than expected. Some explanations for this can be addressed. First, it is reported in literature (Tardieu et al., 1992; Amato and Ritchie, 2002) that the uneven distribution of roots can have significant influence on pressure head gradients in the neighborhood of roots and, hence, on water availability. Therefore, to transform experimentally observed root densities into mean distance between roots, information about the root system architecture should be considered. Second, partial root–soil contact may decrease the roots’ effectiveness in taking up water (Van Noordwijk et al., 1993). Third, the chosen value of h_{lim} (-150 m) may be too negative. Divergent values are reported in the literature: Adeoye and Rawlins (1981) measured -120 m in young maize (*Zea mays* L.) plants; Liu et al. (2005a) observed -250 m in soybean [*Glycine max* (L.) Merr.], while Liu et al. (2005b) found -80 m in

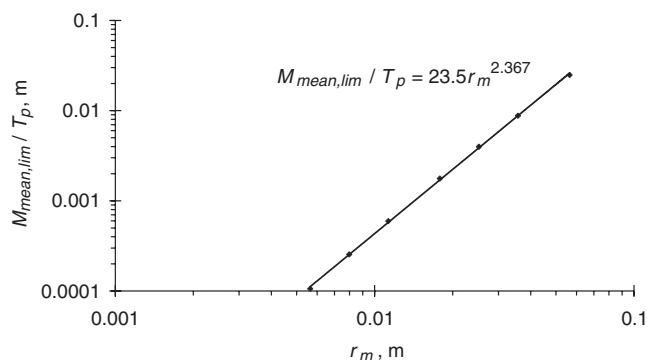


Fig. 11. Ratio between mean matric flux potential at first occurrence of limiting hydraulic conditions and potential transpiration rate ($M_{\text{mean,lim}}/T_p$) as a function of half-distance between roots (r_m).

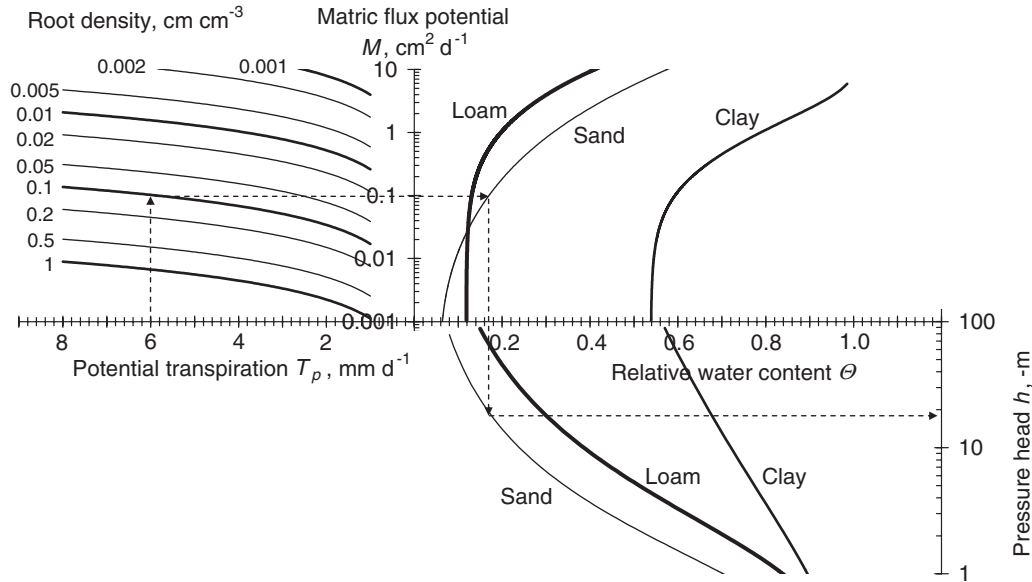


Fig. 12. Nomogram for the determination of pressure head at first occurrence of limiting hydraulic conditions ($h_{\text{mean,lim}}$) and relative water content at first occurrence of limiting hydraulic conditions ($\theta_{\text{mean,lim}}$) for three soils from transpiration and root density data. Dotted arrow lines illustrate nomogram use for a high transpiration rate (6 mm d^{-1}) and a medium root density (0.1 m m^{-3}) for the sand soil.

potato (*Solanum tuberosum* L.). Sensitivity for this value may be an object of future research.

Comparison of Simulation Results to Analytical Solutions

The simulation results show that, at the first occurrence of limiting head at the root surface when transpiration rate is still equal to potential rate but about to drop below that, matric flux potential as a function of the space coordinate r is in good approximation independent of soil characteristics (Fig. 13). The difference between the three soils can be expressed as the relative r -weighted root mean square difference. Comparing M for clay and loam to sand, calculated values are in the range of 0.8 to 3.0%. The weighted mean water content corresponds to the water content in the root zone at some location between $0.53r_m$ and $0.57r_m$ for most cases, slightly depending on soil type, transpiration rate, and root density (Table 3). Further analysis of these values shows a mean of 0.56, a standard deviation of 0.06, and a median of 0.53.

Analytical solutions for root water extraction for specific boundary conditions are reported in the literature, accompanied by simplifying assumptions about the boundary conditions and K - h or D - θ relations. For a soil with constant diffusivity D , infinite radial extent, and for large values of t , a solution to the differential Eq. [10] was presented by Philip (1957), analogous to the Carslaw and Jaeger (1946) solution for the process of thermal conduction:

$$\theta_i - \theta_{\text{mean}} = \frac{r_m^2 T_p}{4zD} \left[\ln \left(\frac{4Dt}{r_{\text{mean}}^2} \right) - \gamma \right] \quad [29]$$

assuming that the average water content θ_{mean} is found at a distance r_{mean} . In this equation, θ_i is the initial water content and γ is the Euler-Mascheroni constant ($\gamma \sim 0.5772$).

Defining $t = t_{\text{lim}}$ as the moment of first occurrence of h_{lim} at the root surface, this equation can be rewritten to express r_{mean} as a function of θ_{mean} at t_{lim} :

$$r_{\text{mean}} = 2\sqrt{Dt_{\text{lim}}} \exp \left[\frac{2zD}{r_m^2 T_p} (\theta_{\text{mean}} - \theta_i) - \frac{\gamma}{2} \right] \quad [30]$$

The average water content can be determined from the r -weighted integral of the $\theta(r)$ profile over r according to Eq. [26] as follows:

$$\theta_{\text{mean}} - \theta_i = \frac{r_m^2 T_p}{4zD} \left[\gamma - \ln(4Dt_{\text{lim}}) + r_m^2 (2 \ln r_m - 1) - r_0^2 (2 \ln r_0 - 1) \right] \quad [31]$$

Substitution of Eq. [31] into [30] yields

$$r_{\text{mean}} = \left[\frac{r_m^2}{r_0^2} \right]^{1/(r_m^2 - r_0^2)} \exp \left(-\frac{1}{2} \right) \quad [32]$$

which, for $r_m \gg r_0$ reduces to

$$r_{\text{mean}} = r_m \exp \left(-\frac{1}{2} \right) \approx 0.6065 r_m \quad [33]$$

indicating that, under the assumptions made, the mean value of θ corresponds to θ at a distance of $0.6065r_m$ from the axial center.

For each scenario of transpiration rate and rooting density, and while the transpiration rate matches potential transpiration rate, flux density at the root surface is independent of soil characteristics. Numerical simulation results show that this independency can be extended to the whole root sphere for the three evaluated soils (Fig. 13). These results also show that the weighted mean water content corresponds to the water content in

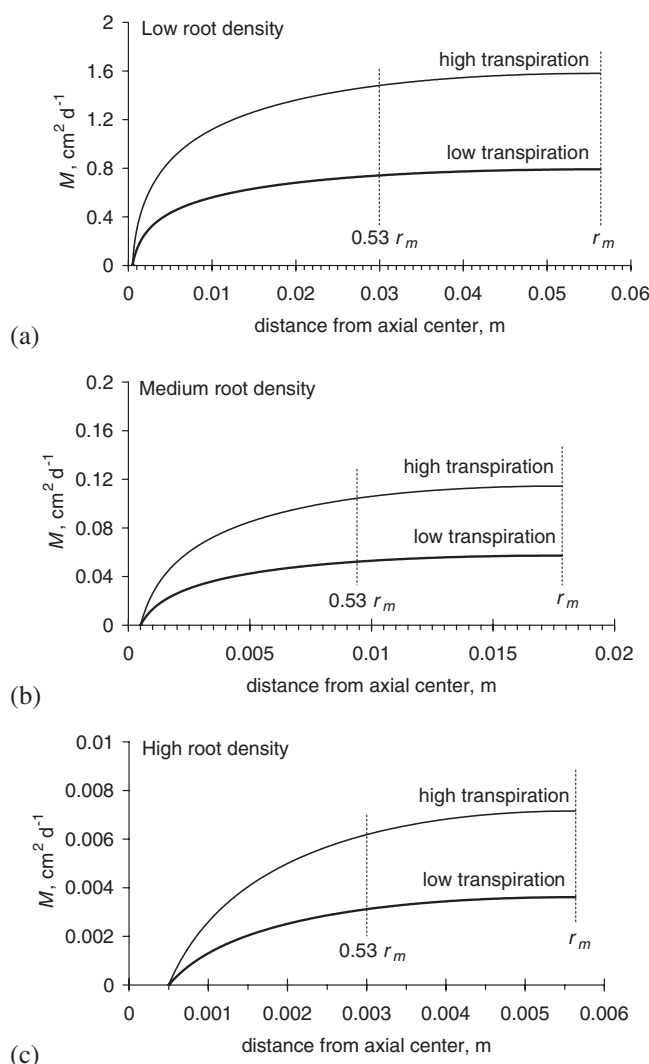


Fig. 13. Matric flux potential (M) at first occurrence of limiting hydraulic conditions as a function of distance from axial center for low, medium, and high root density at low and high potential transpiration (lines for three soils coincide at this resolution); r_m is the radius of the root extraction zone or the half-distance between roots.

the root zone at some location between $0.53r_m$ and $0.57r_m$ for most cases, slightly depending on soil type, transpiration rate, and root density (Table 3).

The simulation-derived values in Table 3 are in rough agreement with Eq. [33], except for some values for the high-root-density scenarios. The assumption of a constant diffusivity causes analytical $\theta(r)$ profiles to be less depleted close to the root surface and, consequently,

Table 3. Fraction of the half-distance between roots (r_m) at which volumetric water content (θ) matches water content at first occurrence of limiting hydraulic conditions ($\theta_{\text{mean,lim}}$) in numerical simulations.

Transpiration	Root density	Sand	Clay	Loam
Low	Low	0.5365	0.5340	0.5469
	Medium	0.5316	0.5478	0.5769
	High	0.5533	0.6199	0.7575
High	Low	0.5314	0.5314	0.5340
	Medium	0.5275	0.5396	0.5478
	High	0.5420	0.5571	0.5054

$\theta_{\text{mean,lim}}$ to occur at a greater distance from that surface than in the numerical simulations. For the high-root-density scenarios, it is probably not realistic to expect Eq. [33] to be valid, as one of the assumptions is an infinite extension of the root zone.

If an analytical expression for $M(r)$ could be found, it would allow the calculation of M at the distance of mean θ , providing a good estimation of $\theta_{\text{mean,lim}}$ or $h_{\text{mean,lim}}$. The continuity equation for one-dimensional, axisymmetric flow (see also Eq. [4]) is

$$\frac{\partial \theta}{\partial t} = -\frac{q}{r} - \frac{\partial q}{\partial r} \quad [34]$$

From mass conservation, it follows that

$$\frac{\partial \theta_{\text{mean}}}{\partial t} = -\frac{T_p}{z[1 - (r_0/r_m)^2]} \quad [35]$$

Ignoring the root volume (assuming $r_m \gg r_0$), Eq. [35] reduces to

$$\frac{\partial \theta_{\text{mean}}}{\partial t} = -\frac{T_p}{z} \quad [36]$$

To solve Eq. [34], an assumption has to be made about the behavior of $d\theta/dt$ as a function of r . Figure 14 shows $h-r$ and $\theta-r$ profiles from numerical simulations with clay soil, medium root density, and a high transpiration rate. In this figure, $h-r$ profiles show a rather sharp increase of dh/dr near the root surface up to the moment of first occurrence of h_{lim} , whereas $\theta-r$ profiles are much

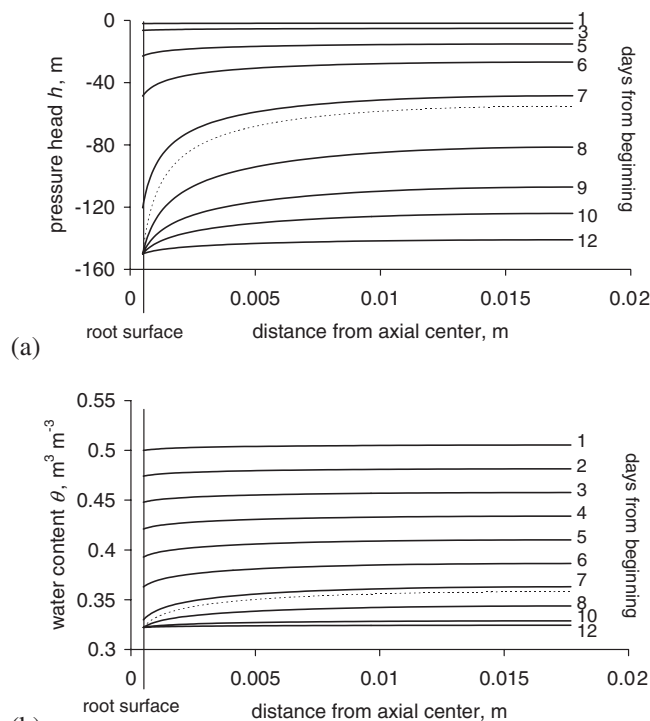


Fig. 14. (a) Pressure head and (b) water content as a function of distance from axial center at several times after start of water extraction (clay soil, medium root density, high transpiration rate). Dotted lines indicate pressure head or water content profile at the moment of first occurrence of limiting hydraulic conditions at the root surface.

more parallel with time, suggesting that $d\theta/dt$ is independent of r and constant during the period in which transpiration is equal to its potential rate. We assume that as long as and up to the moment the transpiration is (still) potential, the profiles of the soil hydraulic properties are in equilibrium with a constant uptake rate at the root surface. De Willigen and Van Noordwijk (1987) argued that, in the case of a roughly constant D , this implies that water content decreases at equal rates independent of the distance; corresponding matric flux potential profiles show a parallel downward shift with time (for studies using the same assumption, see, e.g., De Willigen and Van Noordwijk, 1987; Heinen, 2001; Raats, 2006). Therefore Eq. [36] can be generalized for any r :

$$\frac{\partial \theta}{\partial t} = -\frac{T_p}{z} \quad [37]$$

Equation [36] can be combined with Eq. [34] to yield the following second-order differential equation:

$$\frac{T_p}{z} = -\frac{q}{r} - \frac{\partial q}{\partial r} = \frac{\partial M_{h_0}}{r \partial r} + \frac{\partial^2 M_{h_0}}{\partial r^2} \quad [38]$$

for which the following general solution is found:

$$M_{h_0} = \frac{T_p}{4z} r^2 + C_1 \ln r + C_2 \quad [39]$$

where C_1 and C_2 are integration constants. This solution is known as the steady-rate solution. It was presented by Van Noordwijk and De Willigen (1987) and is also used in the model described by Heinen (2001).

Values for C_1 and C_2 can be found assuming the following boundary conditions to be valid at the moment of first occurrence of limiting root water potential:

$$\frac{dM_{h_0}}{dr} = \frac{T_p A_p}{A_r}; \quad r = r_0 \quad [40]$$

$$M_{h_0} = 0; \quad r = r_0 \quad [41]$$

Equation [40] combined with Eq. [9] yields

$$\frac{dM_{h_0}}{dr} = T_p \frac{r_m^2}{2zr_0}; \quad r = r_0 \quad [42]$$

Combining these boundary conditions with Eq. [39] yields

$$C_1 = \frac{T_p}{2z} (r_m^2 - r_0^2) \quad [43]$$

$$C_2 = \frac{-T_p}{2z} \left[\frac{r_0^2}{2} + (r_m^2 - r_0^2) \ln r_0 \right] \quad [44]$$

and

$$M_{h_0} = \frac{T_p}{2z} \left[\frac{r^2 - r_0^2}{2} + (r_m^2 - r_0^2) \ln \frac{r}{r_0} \right] \quad [45]$$

Equation [45] shows that, under the assumption that $d\theta/dt$ has the same value for all r , $M(r)$ is indeed independent of soil characteristics, as shown in the numerical simulations (Fig. 14). This, combined with the fact that

$\theta_{\text{mean,lim}}$ occurs at a fraction of the root mean distance that is not very sensitive to soil properties, allows estimation of $M_{\text{mean,lim}}$ with Eq. [45] and determination of the corresponding $h_{\text{mean,lim}}$ and $\theta_{\text{mean,lim}}$.

Figure 15 shows the values of $h_{\text{mean,lim}}$ calculated from values of M at first depletion estimated by the analytical model, compared with estimates by the numerical model for the three soils at low root density and high transpiration rate. Among the simulated combinations of root density and transpiration rate, and then especially for the sand soil, this was the only case in which a small discrepancy between numerical and analytical estimates could be observed. For all other cases, discrepancies were much smaller and lines approximately matched the 1:1 line. This confirms that the assumption resulting in Eq. [37] is reasonably well supported by the numerical simulation results.

Given the quality of this analytical result, it can be used to understand why the ratio M/T_p behaves like a power function of r_m (Eq. [28]). The value of M_{mean} is found at the same distance as θ_{mean} (Eq. [27]), which we define here as a distance ar_m from the axial center. Evaluating Eq. [45] at ar_m , neglecting the r_0^2 terms and multiplying by 2 to deal with the fact that critical hydraulic conditions correspond to double transpiration rates (as discussed above), it reduces to

$$\frac{M}{T_p} = \frac{r_m^2}{z} \left[\frac{a}{2} + \ln \frac{ar_m}{r_0} \right] \quad [46]$$

As a is between 0 and 1, and as generally $ar_m \gg r_0$, it is reasonable to neglect the $a/2$ term, further simplifying Eq. [46] to

$$\frac{M}{T_p} = \frac{r_m^2}{z} \ln \frac{ar_m}{r_0} \quad [47]$$

Evaluated for $a = 0.53$ and simulation parameters $r_0 = 0.5$ mm and $z = 0.5$ m, values calculated by Eq. [47] for r_m ranging from 15 to 75 mm fit to Eq. [28] with $p = 18.1$ and $q = 2.277$ ($n = 61$, $r^2 = 0.999$) are similar to the values obtained with numerical simulation.

These results indicate that analytical and numerical methods are in good agreement. As a first approximation, the analytical solution may therefore be used.

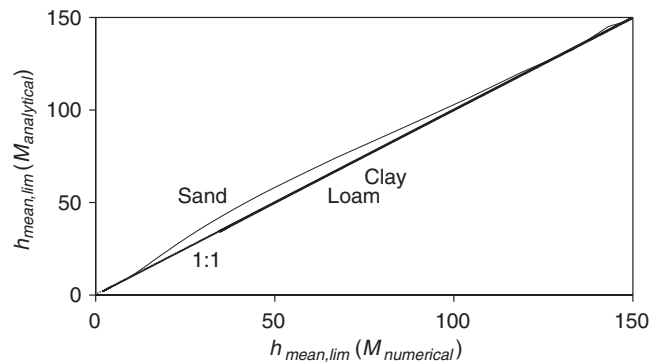


Fig. 15. Pressure head at the first occurrence of limiting root water potential ($h_{\text{mean,lim}}$) calculated from analytical and numerical matric flux potential ($M_{\text{analytical}}$ and $M_{\text{numerical}}$, respectively) for the three soils at low root density and high transpiration rate.

Results from the numerical method do not depend on assumptions about $d\theta/dt$, however, and are therefore preferable. The difference between the numerically estimated distance of mean water content, $0.53r_m$, and its analytical equivalent, $0.6065r_m$, is an indicator of the degree of dependency between $d\theta/dt$ and r . Further investigation of this value for soils with different hydraulic properties, as well as detailed evaluation of the depletion phase after the first occurrence of limiting hydraulic conditions, will be important to provide a physically supported way to estimate soil water availability and actual transpiration rates.

CONCLUSIONS

1. The limiting values (water content or pressure head) of macroscopic root water extraction models are a function of potential transpiration rate, soil hydraulic properties, and root density.
2. In contrast, numerical simulations showed the matric flux potential at first occurrence of limiting hydraulic conditions ($M_{\text{mean,lim}}$) to be independent of soil type and to depend only on potential transpiration rate and root density. Analytically, it was shown that this can be explained if water depletion $d\theta/dt$ has the same value within the entire root sphere. In this case, $M(r)$ is a soil-independent function of transpiration rate and root density.
3. Analytically, mean soil water content is shown to be expected to occur at a distance from the axial center 0.6065 times the half-distance between roots (r_m). Numerical simulations showed that, in fact, this distance varies according to soil type, root density, and transpiration rate. For most simulations, however, the value was around 0.53.
4. Combining the above two conclusions, we suggest that mean soil water content and pressure head at first occurrence of limiting hydraulic conditions ($h_{\text{mean,lim}}$ and $\theta_{\text{mean,lim}}$) can be calculated from the matric flux potential at $0.53 r_m$, which is a function of transpiration rate and root density and which depends on a soil-specific $K(h)$ function.
5. Thus, either the described numerical model, nomograms as depicted in Fig. 12, or the analytical solutions can be used for a comprehensive determination of $h_{\text{mean,lim}}$ and $\theta_{\text{mean,lim}}$.

ACKNOWLEDGMENTS

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